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# Dae Yeon Kim Ph.D.

Assistant Professor, Chemical and Biomolecular Engineering, Chonnam National University
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## EDUCATION

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| **Ph.D.** | **Seoul National University**, Seoul, Korea<br>School of Chemical and Biological Engineering<br>Thesis: "A study on particle deposition and clogging dynamics of colloidal suspensions in complex flow"<br>(Advisor: Prof. Kyung Hyun Ahn) | Aug 2023 |
| **B.S.** | **Sogang University**, Seoul, Korea<br>Department of Chemical and Biomolecular Engineering<br>Graduation honors, *Summa Cum Laude* | Feb 2017 |

## PROFESSIONAL EXPERIENCES

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| **Assistant Professor** | Department of Chemical and Biomolecular Engineering<br>Chonnam National University, Yeosu-si, Jeonnam-Gwangju, Korea | Mar 2026 – Present |
| **Postdoctoral Scholar** | Department of Chemical Engineering<br>Stanford University, Stanford, CA, USA<br>(Advisor: Prof. Sho Takatori) | Feb 2025 – Jan 2026 |
| **Postdoctoral Scholar** | Department of Chemical Engineering<br>University of California, Santa Barbara, Santa Barbara, CA, USA<br>(Advisor: Prof. Sho Takatori) | Mar 2024 – Jan 2025 |
| **Postdoctoral Researcher** | Center for Coating Materials and Processing<br>Seoul National University, Seoul, Korea<br>(Advisor: Prof. Kyung Hyun Ahn) | Sep 2023 – Jan 2024 |

## RESEARCH EXPERIENCES

### Research Skills

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| Microfluidics | • Design and fabrication of microchannel<br>• Fluid visualization and PIV<br>• Pressure drop measurement |
| Rheometry | • Rotational rheometer (ARES, AR-G2, DHR)<br>• Extensional rheometer (Capillary Breakup Extensional Rheometer)<br>• Passive/Active microrheology (Optical tweezer) |
| Capillary breakup | • Measurement of elongational property of complex fluids (polymer solution, colloidal suspension)<br>• Direct visualization of thinning filament surface via high-speed camera |
| Optical tweezer | • Measurement of inter-particle interaction (e.g. Hydrodynamic interaction)<br>• Microrheology of soft matter |
| Chemical labeling | • Soft vesicle labeling (SUV, GUV)<br>• Colloids and bacteria labeling |
| Bacterial culturing | • Characterization and labeling<br>• Quantitative imaging analysis |

Analysis tool
- Image processing and quantitative analysis (MATLAB, Python)
- CFD simulation (COMSOL)

### Research Interests

Complex materials and soft matter physics
- Colloidal suspension rheology
- Fluid-mediated interactions between colloids
- Active matter rheology and control
- Colloidal fouling onto a membrane surface (complex interactions between foulants and membrane)
- Flow instability of viscoelastic fluid

Heterogeneity control of industrial slurry (e.g. lithium-ion batteries, fuel cells, CMP slurry)
- Stability control of colloidal slurry
- Characterization of slurry for quality control

Data-driven machine learning classification
- Characterization of complex fluid
- Clogging and fouling prediction

Image-based rheological properties measurement
- Discovery of constitutive relations for complex fluids
- Dynamics of soft materials with active forcing

## PUBLICATIONS

### Peer-reviewed Journal Articles

* equal contribution † co-corresponding

14. **Dae Yeon Kim**, Colloidal hydrodynamic interactions: from bulk rheometry to rotation-based optical tweezer measurements, *Korea-Australia Rheology Journal*, **2026**, published online. [Link]

13. Haegyo Jeong, Minsun Hwang, **Dae Yeon Kim**†, Seon Yeop Jung†, Influence of fluid viscosity and cavity length on particle transport and deposition in expansion-contraction microchannels, *Korea-Australia Rheology Journal*, **2026**, 38 (3), 335–354. [Link]

12. **Dae Yeon Kim***, Sachit G. Nagella*, Saksham Malik, Nayeon Park, Jaewook Nam, Eric S.G. Shaqfeh, Sho C. Takatori, Colloidal hydrodynamic interactions in viscoelastic fluids, *Soft Matter*, **2025**, 21, 8035–8048. [Link]

11. Minsun Hwang, **Dae Yeon Kim**, Kyung Hyun Ahn, Sung Sik Lee, Seon Yeop Jung, Particle deposition in colloid-polymer suspensions: Complex interplay between flow characteristics and polymer interactions, *Physics of Fluids*, **2025**, 37, 073389. [Link]

10. **Dae Yeon Kim***, Sachit G. Nagella*, Kyu Hwan Choi, Sho C. Takatori, Direct experimental measurement of many-body hydrodynamic interactions with optical tweezers, *Physical Review Fluids*, **2025**, 10, 064301. [Link]

9. Woobin Yi*, **Dae Yeon Kim***, Howon Jin, Sangwoong Yoon, Kyung Hyun Ahn, Early detection of pore clogging in microfluidic systems with 3D convolutional neural network, *Separation and Purification Technology*, **2025**, 359, 130428. [Link]

8. Young Jin Lee, **Dae Yeon Kim**, Kyung Hyun Ahn, Particulate fouling simulation in unit micropore using a hydrodynamically coupled Lagrangian framework, *Desalination*, **2025**, 593, 118255. [Link]

7. Minsun Hwang, **Dae Yeon Kim**, Seon Yeop Jung, Kyung Hyun Ahn, Effect of fluid viscosity on the deposition of colloidal particles in an expansion-contraction microchannel flow, *Nihon Reoroji Gakkaishi*, **2024**, 52 (4), 231–240. [Link]

6. Wonsik Shin, Joung Sook Hong, **Dae Yeon Kim**, Si Yoon Kim, Kyu Hyun, Jun Dong Park, Kyung Hyun Ahn, Effect of cellulose nanocrystals on the emulsion stability and rheological properties of microalgal Pickering emulsions, *Algal Research*, **2024**, 83, 103731. [Link]

5. Young Jin Lee, Howon Jin, **Dae Yeon Kim**, Seunghoon Kang, Kyung Hyun Ahn, Lagrangian-based simulation method using constrained Stokesian dynamics for particulate flows in microchannel, *Journal of Fluid Mechanics*, **2024**, 990, A3. [Link]

4. **Dae Yeon Kim**, Seon Yeop Jung, Young Jin Lee, Howon Jin, Kyung Hyun Ahn, From successive deposition to clogging of poly(styrene) particles in the cross-flow through a T-shaped microchannel, *Separation and Purification Technology*, **2023**, 322, 124263. [Link]

3. **Dae Yeon Kim**, Seon Yeop Jung, Young Jin Lee, Kyung Hyun Ahn, Effect of colloidal interactions and hydrodynamic stress on particle deposition in a single micropore, *Langmuir*, **2022**, 38 (19), 6013–6022. [Link]

2. **Dae Yeon Kim**, Kyung Hyun Ahn, Effect of operating conditions on the sessile drop oscillation of PEO solution in the capillary thinning process, *Nihon Reoroji Gakkaishi*, **2021**, 49 (2), 127–133. [Link]

1. Youngseok Kim, **Dae Yeon Kim**, Joung Sook Hong, Kyung Hyun Ahn, Agglomerate breakup of destabilized polystyrene particles under a cross-channel planar extensional flow, *Langmuir*, **2018**, 34 (38), 11454–11463. [Link]

### Submitted/in-progress

3. **Dae Yeon Kim**, Youngseok Kim, Kyung Hyun Ahn, Colloidal clogging of a model membrane pore in Newtonian and viscoelastic media, *in preparation*.

2. **Dae Yeon Kim**, Free-vibration mechanism and viscoelastic damping of sessile-drop oscillation in capillary thinning, *submitted*.

1. **Dae Yeon Kim**, Izzie Allydice, Sho C. Takatori, Colloidal hydrodynamic interaction in bacterial active bath, *in preparation*.

## PRESENTATIONS

### International Conference

9. Fluid-mediated rotational coupling and hydrodynamic interactions in colloidal particles using optical trapping, *APS Global Physics Summit (APS March Meeting 2025)*. 16 – 21 Mar 2025, Anaheim, CA, USA.

8. Successive deposition of poly(styrene) particles in cross-flow through T-shaped microchannel, *The 19th International Congress on Rheology (ICR 2023)*. 29 July – 4 Aug 2023, Athens, Greece.

7. Effect of salt concentration and flow rate on clogging dynamics at the single micro-pore. *The 8th Pacific Rim Conference on Rheology (PRCR 2023)*. 15 – 19 May 2023, Vancouver, BC, Canada.

6. Effect of salt concentration and flow rate on clogging dynamics at the single micro-pore. *The 93rd Annual Meeting of the Society of Rheology (SoR 93rd Annual Meeting)*. 9 – 13 Oct 2022, Chicago, IL, USA.

5. Effect of salt concentration and flow rate on clogging dynamics at the single micro-pore. *11th Korea-Australia Rheology Conference (KARC 2021)*. 24 – 25 Nov 2021, Virtual.

4. The effect of operating condition and rheological properties of PEO solution on the sessile drop oscillation during elasto-capillary thinning. *International Union of Materials Research Societies-International Conference in Asia (IUMRS-ICA 2020)*. 23 – 26 Feb 2021, Virtual.

3. Effect of rheological properties of PEO solution on the sessile drop oscillation during the early stages of elasto-capillary thinning transition. *The 15th International Workshop for East Asian Young Rheologists (IWEAYR 2020)*. 8 – 10 Jan 2020, Changchun, China.

2. Effect of rheological properties of PEO solution on the sessile drop oscillation during the early stages of elasto-capillary thinning transition. *Annual European Rheology Conference (AERC 2019)*. 8 – 11 Apr 2019, Portoroz, Slovenia.

1. Agglomerate breakup of destabilized polystyrene particles under a cross-channel planar extensional flow. *The 14th International Workshop for East Asian Young Rheologists (IWEAYR 2019)*. 23 – 26 Jan 2019, Nagoya, Japan.

### Others (2023 – Present)

5. Particle Deposition, Pore Clogging, and AI-Assisted Prediction of Membrane Fouling. *KSIEC Special Symposium: AX in the Carbon-Neutral Era, Korean Society of Industrial and Engineering Chemistry*. 10 – 11 Sep 2026, Yeosu, Korea

4. Probing hydrodynamic interactions and rheology in complex fluids using optical laser tweezers. *2026 Annual Meeting of the Korean Society of Rheology*. 7 – 8 May 2026, Seoul, Korea

3. Prediction of membrane fouling via particle transport phenomena and ai-based analysis. *2026 Spring Conference of the Korean Society for Energy*. 15 – 17 Apr 2026, Yeosu, Korea

2. Direct experimental measurement of hydrodynamic interactions using optical tweezers. *Shaqfeh Group Seminar, Stanford University*. May 2025

1. A study on particle deposition and clogging dynamics of colloidal suspensions in complex flow. *EEPL Group Seminar, Dankook University*. Aug 2023

## TEACHING AND MENTORING

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| **Teaching** (S:Spring, F:Fall) | Transport Phenomena (Chonnam National University) | 2026F |
| | Chemical Reaction Engineering II (Chonnam National University) | 2026F |
| | General Chemistry I (Chonnam National University) | 2026S |
| | Chemical Reaction Engineering I (Chonnam National University) | 2026S |
| | Creative Academy: Basic Chemical Engineering (Janghoon High school, Seoul) | Mar 2018 – Jan 2019 |
| **TA** | Transport Phenomena (Seoul National University) | Spring. 2018 |
| | Introduction to Chemical Engineering (Seoul National University) | Fall. 2019 |
| **Mentoring** | 2025 ChemE REU Program<br>Mentee: Izzie Allydice (Stanford '28) | Jul 2025 – Present |
| | EEPL Group, Dankook University<br>Mentee: Minsun Hwang (B.S.–M.S. Integrated Program) | Mar 2022 – Jul 2025 |

## AWARDS AND HONORS

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| **Best Paper Award** | DONGJIN Best Paper Award<br>(supported by DONGJIN SEMICHEM CO., LTD.) | Jul 2022 |
| **Best Presentation Award** | Best Presentation Award of 15th IWEAYR<br>(The 15th International Workshop for East Asian Young Rheologists) | Jan 2020 |
| **Academic Scholarship** | Scholarship for academic excellence<br>(Sogang University) | Sep 2013 – Aug 2016 |

## OTHER ACTIVITIES

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| **Member** | Young Engineers Honor Society (YEHS) | Jan 2014 – Present |
| **Student Leader** | Department of Chemical and Biomolecular Engineering<br>Sogang University, Korea | Jan 2014 – Mar 2014 |

## REFERENCES

| | | |
|---|---|---|
| **Dr. Kyung Hyun Ahn** | Professor,<br>School of Chemical and Biological Engineering,<br>Seoul National University, Seoul, Korea | ahnnet@snu.ac.kr |
| **Dr. Sho Takatori** | Associate Professor,<br>Department of Chemical Engineering,<br>Stanford University, Stanford, CA | stakatori@stanford.edu |
| **Dr. Seon Yeop Jung** | Assistant Professor,<br>Department of Chemical Engineering,<br>Hongik University, Seoul, Korea | syjung@hongik.ac.kr |